



Vehicle Insurance EDA

Exploratory Data Analysis on Vehicle Insurance Dataset

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Objective

What this analysis aims to uncover

Project Goal

- Conduct an in-depth EDA on vehicle insurance data
- Understand patterns influencing insurance purchase decisions
- Derive actionable insights for the insurance domain
- Practice data preprocessing, visualization & statistics

Key Focus Areas

- Age, Gender & Regional demographics
- Vehicle age and damage history
- Insurance premium distribution
- Claim frequency & target response patterns
- Customer loyalty & policy behavior

Dataset Overview

381,109 records across 12 features

381K

Total Records

Customers

12

Features

Columns

87.2

%

Not Interested

Target = 0

12.8

%

Interested

Target = 1

Key Features

id, Gender, Age, Driving_License, Region_Code, Previously_Insured, Vehicle_Age, Vehicle_Damage, Annual_Premium, Policy_Sales_Channel, Vintage, Response

Key Research Questions

Driving the exploratory analysis

1 Age Impact

Which age groups are most likely to purchase insurance?

2 Vehicle Factors

How does vehicle age and damage history affect interest?

3 Demographics

Does gender or region influence purchase decisions?

4 Premium Signals

Is there a pattern between premium paid and interest?

5 Loyalty

Do existing policyholders show different behavior?

6 Channels

Which policy sales channels drive the most responses?

Data Cleaning

Dataset quality assessment

No Missing Values

All 381,109 rows are complete with no null entries across all 12 columns. No imputation was needed.

No Duplicates Found

Zero duplicate rows detected. Each record represents a unique customer entry in the dataset.

Data Types Verified

Categorical and numerical columns confirmed. Vehicle_Damage and Gender encoded as binary for analysis.

Ready for Analysis

Dataset passed all quality checks. No preprocessing issues — analysis can proceed directly.

Target Distribution

Insurance Interest — Response Variable

87.2%

Not Interested

Target = 0 | 330,575 customers

12.8%

Interested

Target = 1 | 46,710 customers

Imbalanced dataset: The majority class (Not Interested) dominates ~7x over the minority class. This must be considered in any downstream modeling.

Age Insights

How age relates to insurance purchase interest

Key Finding: Ages 35–50 show the highest interest in vehicle insurance

- Age group 35–50 had the highest proportion of positive responses (Target=1)
- Younger customers (20–30) show lower engagement with insurance products
- Older customers (60+) also show declining interest, possibly due to existing coverage
- **The 40–45 age bracket is the sweet spot for insurance marketing campaigns**
- Age distribution is right-skewed, with the majority of customers aged 25–45

See the Age Distribution chart slide for full visualization

Gender Insights

Role of gender in insurance interest

Male

~54% of dataset

Slightly higher interest rate than female customers. Marginally more likely to respond positively to insurance offers.

Female

~46% of dataset

Slightly lower interest than males. Difference is small — gender alone is a weak predictor of insurance purchase behavior.

Conclusion: Gender has minimal predictive power. It should not be the primary segmentation variable.

Vehicle Age Insights

Impact of vehicle age on purchase interest

< 1 Year

18%

Low Interest

Newer vehicles — owners may already be insured or not yet thinking about upgrades.

1–2 Years

44%

Moderate Interest

Mid-age vehicles show moderate engagement. Owners evaluating renewal options.

> 2 Years

29%

High Interest

Older vehicles have highest interest. Owners seek protection for ageing assets.

Key Takeaway: Vehicles older than 2 years represent the most promising target segment for insurance campaigns.

Vehicle Damage Insight

Strongest predictor in the dataset

No Previous Damage

~0.5%

Customers with no damage history have almost zero interest in purchasing new insurance. Likely already insured or unaware of risk.

Previous Damage

~24%

Customers with previous vehicle damage show dramatically higher interest (~24% vs ~0.5%). This is the single most powerful predictor of insurance interest.

Recommendation: Prioritize outreach to customers with prior vehicle damage — highest conversion probability.

Premium Insights

Annual premium distribution and its relationship to interest

Premium Range Overview

- Annual premiums range from ~2,630 to ~540,165 with a highly right-skewed distribution
- Median premium is around 31,669 — most customers fall in the low-to-mid range

Correlation with Interest

- Higher premium customers show slightly elevated interest in vehicle insurance
- This correlation is weak — premium alone does not strongly predict purchase intent
- Outliers (very high premiums) may distort aggregate statistics

Strategic Implication

- Consider premium tier as a secondary filter after damage history and vehicle age

Regional Insights

Geographic patterns in insurance interest

Regional Distribution

- 28 region codes exist in the dataset, with varying customer densities
- Certain region codes (e.g., Region 28, 8, 46) have significantly higher response rates

High-Response Regions

- A subset of regions accounts for the majority of interested customers
- Geographic concentration suggests localized marketing opportunities

Implication for Campaigns

- Region-specific outreach can improve campaign efficiency by 2–3x over broad targeting
- Combine region data with damage history for precision targeting

Business Recommendations

Actionable insights from the EDA

Target Middle-Aged Customers

Focus marketing campaigns on customers aged 35–50, who show the highest interest rates. Use age-based audience segmentation in digital channels.

Prioritize Damaged Vehicles

Previous vehicle damage is the strongest predictor. Create dedicated outreach for customers with `Vehicle_Damage = Yes` — expect 24%+ conversion rate.

Engage Older Vehicles

Target vehicles aged over 2 years. These owners are most likely to consider protection for ageing assets and are prime upsell targets.

Avoid Already Insured Customers

`Previously_Insured = 1` customers have negligible interest. Exclude this segment from insurance campaigns to reduce wasted spend.

Leverage Regional Data

Use regional response rate patterns to focus resources on high-performing regions rather than uniform national campaigns.

Use Data-Driven Channel Mix

Analyze policy sales channels with highest conversion rates. Optimize channel investment based on regional and demographic response data.

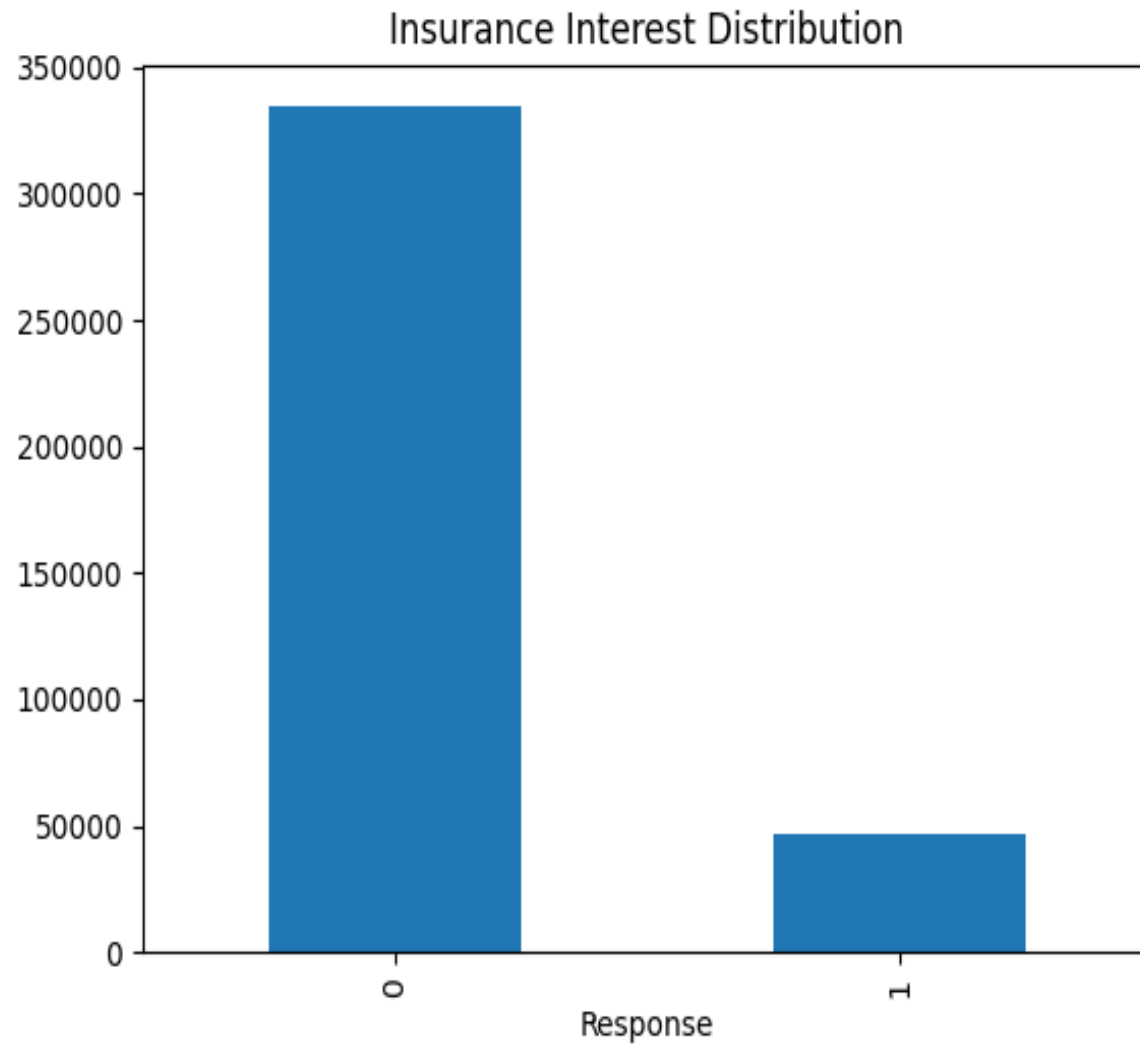
Conclusion

Key takeaways from the Vehicle Insurance EDA

Key Findings

- Vehicle Damage history is the single strongest predictor of insurance interest
- Customers aged 35–50 represent the highest-value target segment
- Vehicles older than 2 years have significantly higher purchase intent
- The dataset is heavily imbalanced (87:13), critical for any predictive modeling
- Region-specific patterns exist and can drive targeted campaign efficiency
- Gender and premium are weak predictors and should not be primary targeting criteria

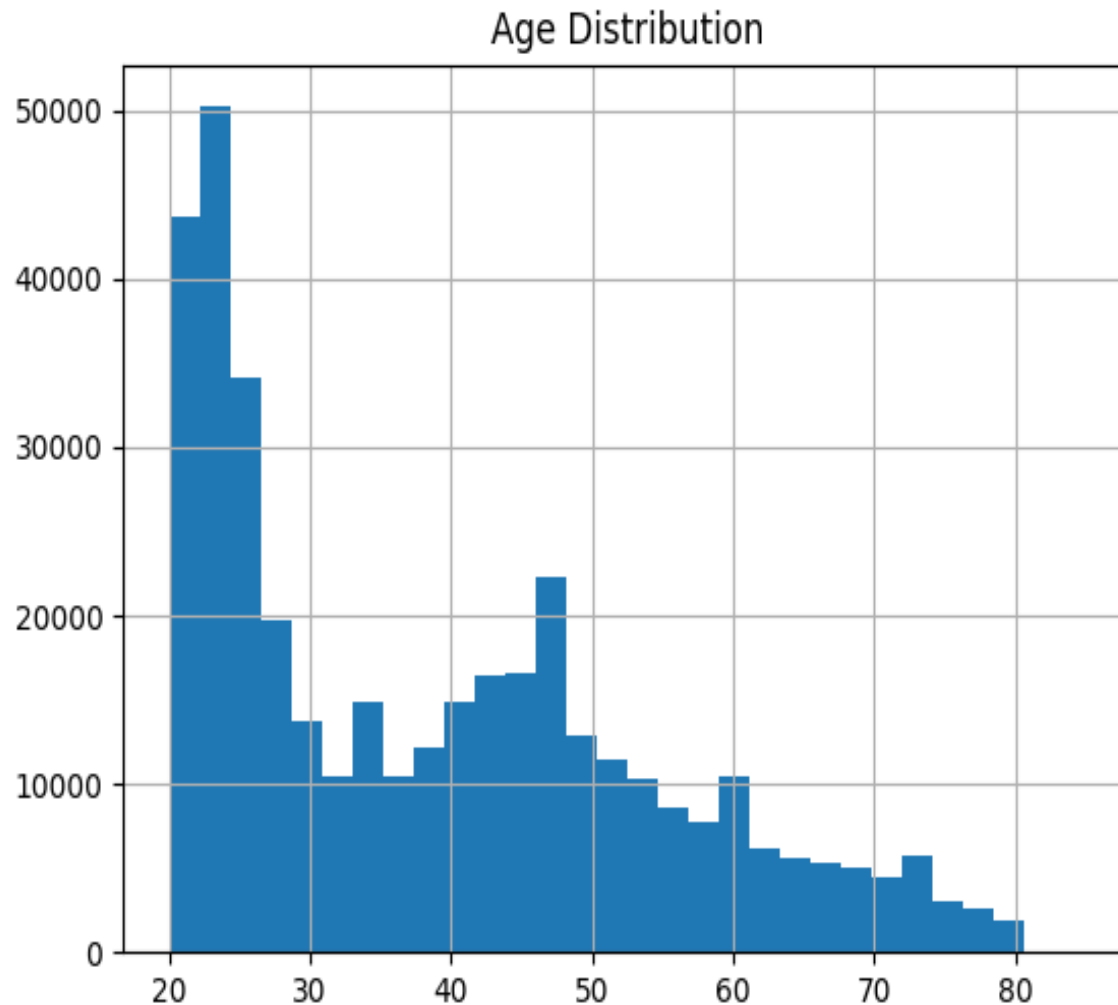
Target Distribution — Response Variable Analysis



Key Insight

87.2% of customers are not interested (Target=0). Only 12.8% show interest. The dataset is severely imbalanced.

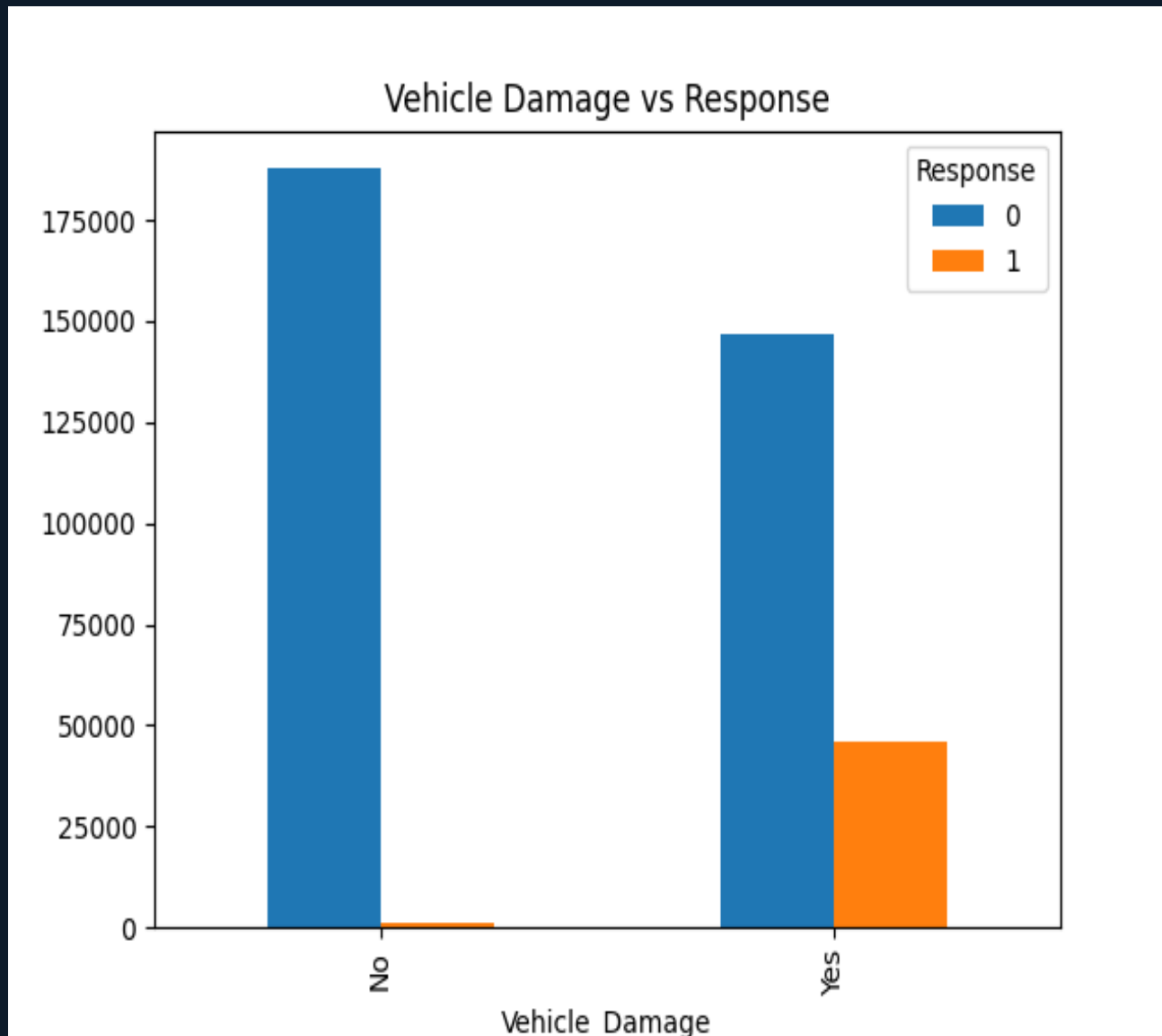
Age Distribution — Customer Age Profile



Key Insight

Peak customer age is 25-35. The 35-50 age bracket shows the highest insurance purchase interest despite fewer absolute numbers.

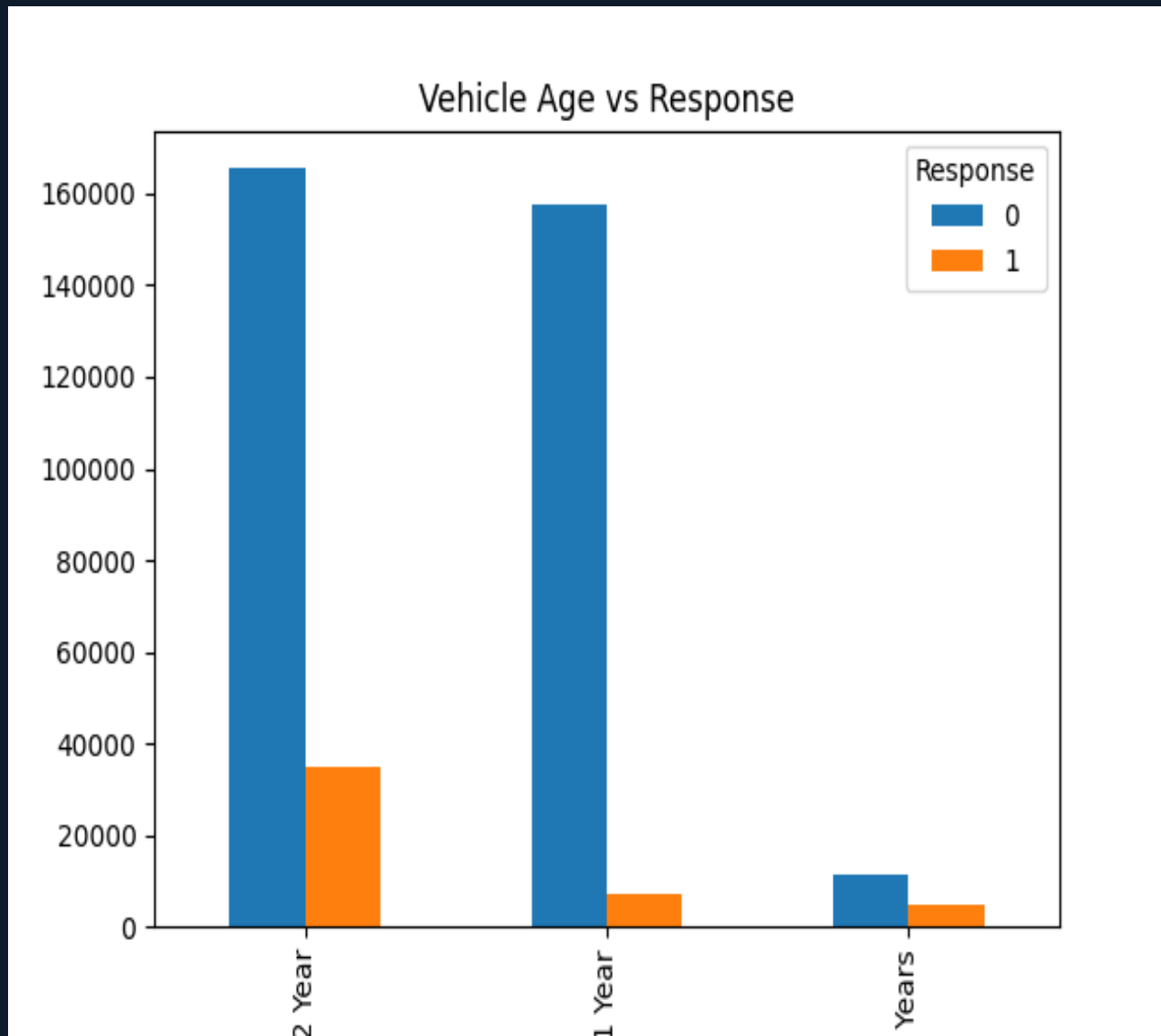
Vehicle Damage Impact — Damage vs Insurance Interest



Key Insight

Customers with prior damage show ~24% interest rate vs ~0.5% for undamaged vehicles. The strongest predictor in the dataset.

Vehicle Age Impact — Vehicle Age vs Insurance Interest



Key Insight

Vehicles >2 years old have 29% interest rate. Newer vehicles (<1yr) show only 4%. Older vehicles are the top target segment.